

1704. Determine if String Halves Are Alike

Easy Topics Companies Hint

You are given a string `s` of even length. Split this string into two halves of equal lengths, and let `a` be the first half and `b` be the second half.

Two strings are **alike** if they have the same number of vowels (`'a'`, `'e'`, `'i'`, `'o'`, `'u'`, `'A'`, `'E'`, `'I'`, `'O'`, `'U'`). Notice that `s` contains uppercase and lowercase letters.

Return `true` if `a` and `b` are **alike**. Otherwise, return `false`.

Example 1:

Input: `s = "book"`
Output: `true`
Explanation: `a = "bo"` and `b = "ok"`. `a` has 1 vowel and `b` has 1 vowel. Therefore, they are alike.

Example 2:

Input: `s = "textbook"`
Output: `false`
Explanation: `a = "text"` and `b = "book"`. `a` has 1 vowel whereas `b` has 2. Therefore, they are not alike. Notice that the vowel `o` is counted twice.

Constraints:

- `2 <= s.length <= 1000`
- `s.length` is even.
- `s` consists of **uppercase and lowercase** letters.

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```
1 class Solution {
2     public boolean halvesAreAlike(String s) {
3         String vowels = "aeiouAEIOU";
4         int n = s.length() / 2;
5
6         int countA = 0;
7         int countB = 0;
8
9         for (int i = 0; i < n; i++) {
10             if (vowels.indexOf(s.charAt(i)) != -1) {
11                 countA++;
12             }
13         }
14
15         for (int i = n; i < s.length(); i++) {
16             if (vowels.indexOf(s.charAt(i)) != -1) {
17                 countB++;
18             }
19         }
20         return countA == countB;
21     }
22 }
```

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☒ Testcase [Test Result](#)

You must run your code first

Difficulty: 1159



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Lapindromes

Lapindrome is defined as a string which when split in the middle, gives two halves having the same characters and same frequency of each character. If there are odd number of characters in the string, we ignore the middle character and check for lapindrome. For example **gaga** is a lapindrome, since the two halves **ga** and **ga** have the same characters with same frequency. Also, **abccab**, **rotor** and **xyzyx** are a few examples of lapindromes. Note that **abbaab** is NOT a lapindrome. The two halves contain the same characters but their frequencies do not match.

Your task is simple. Given a string, you need to tell if it is a lapindrome.

Input:

First line of input contains a single integer **T**, the number of test cases.

Each test is a single line containing a string **S** composed of only lowercase English alphabet.

Output:

For each test case, output on a separate line: "YES" if the string is a lapindrome and "NO" if it is not.

Constraints:

- $1 \leq T \leq 100$
- $2 \leq |S| \leq 1000$, where $|S|$ denotes the length of **S**

Sample 1:

Input	Output
6	YES
gaga	NO
abcde	YES
rotor	YES
xyzyx	NO
abbaab	NO

```
1 import java.util.*;
2
3 class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int T = sc.nextInt();
8
9         while (T-- > 0) {
10             String s = sc.next();
11
12             int n = s.length();
13             int[] freq = new int[26];
14
15             // Count characters in the first half
16             for (int i = 0; i < n / 2; i++) {
17                 freq[s.charAt(i) - 'a']++;
18             }
19
20             // Count characters in the second half
21             for (int i = (n + 1) / 2; i < n; i++) {
22                 freq[s.charAt(i) - 'a']--;
23             }
24
25             boolean isLapindrome = true;
26
27             for (int count : freq) {
28                 if (count != 0) {
29                     isLapindrome = false;
30                     break;
31                 }
32             }
33         }
34     }
```

Test against Custom Input

```
6
gaga
abcde
rotor
```

Visualize Code

Run

Submit

Java



```
37     sc.close();
38 }
39 }
40
```

Test against Custom Input

```
6
gaga
abcde
rotor
```

Status: Successfully executed

Time: 0.0800 secs
Memory: 42.156 Mb

Sample Input

```
6
gaga
abcde
rotor
xyzxy
abbaab
ababc
```

Your Output

```
YES
NO
YES
YES
NO
NO
```